Erwin Quanten

Staff AI/ML Engineer – Python, Machine Learning, LLM, AWS
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PROFESSIONAL SUMMARY

Staff Software Engineer specializing in **Python 3.7–3.12**, **AI/ML platforms**, and **LLM-enabled cloud applications** across managed cloud services, digital transformation, chemical manufacturing, and enterprise data operations. Strong background building **FastAPI 0.95–0.115**, **Django 2.2–4.2**, **PyTorch 1.8–2.3**, **TensorFlow 2.x**, **scikit-learn 0.22–1.5**, **LangChain 0.1–0.2**, **LlamaIndex 0.9–0.11**, and **AWS** systems with production-grade APIs, RAG workflows, vector search, model monitoring, CI/CD, and secure data pipelines that improve reliability, reduce manual work, and support measurable business decisions for engineering, operations, and customer-facing teams.

SKILLS

- **Languages: Python 3.7–3.12**, SQL, TypeScript, JavaScript, Bash
- **AI/ML: PyTorch 1.8–2.3**, **TensorFlow 2.x**, **scikit-learn 0.22–1.5**, XGBoost 1.x–2.x, LightGBM 3.x–4.x, Pandas 1.x–2.x, NumPy 1.18–2.x
- **LLM / GenAI: LangChain 0.1–0.2**, **LlamaIndex 0.9–0.11**, OpenAI API, Azure OpenAI, Hugging Face Transformers 4.x, RAG, embeddings, prompt engineering, evaluation pipelines
- **Backend: FastAPI 0.95–0.115**, **Django 2.2–4.2**, Flask 1.x–3.x, Pydantic 1.10–2.x, Celery 4.x–5.x, REST APIs, GraphQL
- **Vector & Data Stores:** PostgreSQL 12–16, pgvector 0.5–0.7, Pinecone, FAISS, Redis 5–7, Elasticsearch 7–8, MongoDB 4–6
- **Cloud & DevOps: AWS**, ECS, Lambda, S3, RDS, SageMaker, CloudWatch, Docker, Kubernetes 1.22–1.30, Terraform 1.x, GitHub Actions, GitLab CI
- **MLOps:** MLflow 1.x–2.x, model registry, feature pipelines, experiment tracking, monitoring, drift detection, batch inference, real-time inference
- **Architecture:** Microservices, event-driven systems, secure API design, data pipelines, distributed processing, observability, cost optimization

EXPERIENCE

Lightedge — Lead Software Engineer | **May2022 – Apr 2026**
Remote/ Full time | *Houston, TX*

- Managed cloud services and infrastructure automation platforms were enhanced with **Python 3.10–3.12**, **FastAPI 0.95–0.115**, and **AWS** to support AI-assisted operational workflows for internal engineering and customer support teams.
- Designed a production **RAG-based knowledge assistant** using **LangChain 0.1–0.2**, **OpenAI API**, **pgvector 0.5–0.7**, and PostgreSQL to reduce manual incident investigation time by **32%**.
- Built secure **LLM orchestration APIs** with FastAPI, Pydantic 2.x, Redis 7, and structured prompt templates to standardize ticket triage, log summarization, and infrastructure recommendation workflows.
- Implemented embedding pipelines for runbooks, support tickets, architecture documents, and monitoring alerts, improving semantic search accuracy by **27%** across cloud operations content.
- Migrated legacy Python services from **Python 3.8** to **Python 3.11**, resolving dependency conflicts, async runtime issues, and packaging errors through staged testing and CI validation.
- Improved model-serving reliability by containerizing inference services with **Docker**, deploying workloads on **Kubernetes 1.24–1.29**, and adding health checks, retry policies, and resource limits.
- Reduced API latency by **24%** by optimizing async FastAPI endpoints, PostgreSQL indexes, Redis cache usage, and vector similarity queries under production traffic.
- Developed **MLflow 2.x** experiment tracking workflows for classification models, LLM prompt tests, and retrieval evaluation metrics to make model decisions auditable and repeatable.
- Fixed recurring production issues where LLM responses produced inconsistent JSON by adding schema validation, retry parsing, guardrails, and fallback responses through **Pydantic** models.
- Integrated **AWS CloudWatch**, OpenTelemetry, and structured logging to monitor inference errors, token consumption, API failures, and degraded RAG retrieval quality.
- Implemented CI/CD pipelines with GitHub Actions, Terraform 1.x, Docker build caching, unit tests, security scanning, and blue-green deployment checks for AI-enabled backend services.
- Collaborated with product, DevOps, and support teams to translate operational pain points into AI features that improved troubleshooting speed, reduced repetitive support analysis, and increased platform reliability.

Syscom Digital — Staff Software Engineer **Nov 2017 – Apr 2022**
Remote/ Full time *Bucharest, Romania*

- Digital transformation platforms were delivered with **Python 3.6–3.9**, **Django 2.2–3.2**, **Flask 1.x–2.x**, and early **FastAPI 0.60–0.75** services for enterprise analytics and automation use cases.
- Built NLP classification services with **Hugging Face Transformers 2.x–4.x**, **PyTorch 1.4–1.10**, and scikit-learn to categorize customer messages, documents, and operational requests.
- Improved document processing accuracy by **29%** by combining rule-based validation, TF-IDF baselines, transformer embeddings, and human-reviewed feedback loops.
- Designed REST APIs for ML predictions, batch scoring, and analytics dashboards using Django REST Framework, Flask, PostgreSQL, Redis, and Celery-based background workers.
- Migrated older Python 2.7 and Django 1.11 services to **Python 3.8** and **Django 3.2**, fixing Unicode issues, dependency conflicts, ORM changes, and deprecated middleware behavior.
- Reduced manual reporting work by **34%** through automated ETL jobs built with Pandas, SQLAlchemy, scheduled Celery tasks, and data validation checks.
- Implemented model retraining workflows using MLflow 1.x, GitLab CI, Docker, and controlled promotion stages to prevent untested models from reaching production APIs.
- Resolved production memory leaks in document ingestion workers by profiling Pandas transformations, optimizing chunk sizes, and moving large processing tasks into isolated Celery queues.
- Created feature engineering pipelines for customer segmentation, churn indicators, anomaly detection, and forecasting models using scikit-learn, XGBoost, Pandas, and PostgreSQL.
- Worked with architects and business stakeholders to convert vague AI ideas into realistic ML features with measurable acceptance criteria, monitoring rules, and release-safe delivery plans.

Tessenderlo Group — Senior Software Engineer **Aug 2015 – Oct 2017**
Remote/ Full time *Brussel Metropolitan Area*

- Chemical manufacturing and supply chain analytics systems were developed with **Python 3.4–3.6**, **Django 1.8–1.11**, **Flask 0.10–0.12**, Pandas 0.18–0.20, and PostgreSQL.
- Built forecasting models with **scikit-learn 0.17–0.19**, NumPy, and Pandas to support demand planning, inventory analysis, and production trend visibility.
- Improved forecast preparation efficiency by **23%** by replacing spreadsheet-heavy workflows with Python data pipelines, scheduled jobs, and automated validation reports.
- Created internal dashboards and APIs that exposed production KPIs, quality metrics, and logistics indicators to planning teams through secure Django-based applications.
- Reduced recurring data reconciliation issues by **28%** by implementing stronger ETL validation, duplicate detection, schema checks, and exception reporting.
- Migrated legacy reporting scripts from Python 2.7 to **Python 3.5**, solving encoding problems, library incompatibilities, and outdated database connector issues.
- Implemented anomaly detection workflows for production and logistics data using statistical baselines, scikit-learn models, and alert rules reviewed by domain experts.
- Resolved slow reporting queries by redesigning SQL joins, adding PostgreSQL indexes, caching repeated calculations, and moving heavy processing into scheduled background jobs.
- Worked flexibly across backend engineering, data analysis, stakeholder workshops, and production-support activities to deliver practical software improvements in a manufacturing environment.

Tessenderlo Group — Software Engineer **Sep 2011 – Jul 2015**
Hybrid/ Full time *Brussel Metropolitan Area*

- Enterprise manufacturing applications were maintained with **Python 2.7–3.3**, **Django 1.4–1.7**, Flask 0.9–0.10, Celery 3.x, PostgreSQL, and Linux-based deployment scripts.
- Developed data automation tools that collected production, inventory, and procurement data from multiple systems and generated reliable operational reports for business teams.
- Reduced manual data preparation effort by **31%** by replacing spreadsheet macros with Python ETL scripts, SQL queries, and scheduled validation workflows.
- Built early predictive reporting components with NumPy, Pandas 0.13–0.16, and basic statistical models to identify supply chain exceptions and planning inconsistencies.
- Migrated internal Django applications from older framework versions to **Django 1.7**, resolving URL routing changes, ORM warnings, template updates, and deployment configuration issues.

- Fixed recurring background job failures by improving Celery retry handling, logging failed payloads, separating long-running jobs, and adding recovery documentation.
- Improved application reliability by **26%** through better exception handling, database transaction controls, input validation, and automated regression checks.
- Supported cross-functional users by translating manufacturing workflow problems into maintainable Python tools, simple interfaces, and reliable backend services.

Tessenderlo Group — Software Developer **Jun 2009 – Aug 2011**
Hybrid/ Full time *Brussel Metropolitan Area*

- Internal business applications were built with **Python 2.6–2.7**, **Django 1.1–1.3**, SQLAlchemy 0.5–0.6, PostgreSQL, JavaScript, and Linux shell scripting.
- Created reporting utilities for finance, operations, and production teams that reduced repetitive manual data consolidation by **22%**.
- Maintained legacy Python modules, fixed deployment errors, improved SQL query performance, and documented operational steps for support teams.
- Implemented data import scripts with validation rules, error logs, and retry handling to reduce broken records and improve confidence in internal reporting.
- Built a strong foundation in backend development, database design, automation, debugging, and production support within enterprise manufacturing systems.

EDUCATION

KU Leuven — Master’s Degree, Computer Science **Sep 2003 – May 2009**